VOILA: Value-of-Information Guided Fidelity Selection for Cost-Aware Multimodal Question Answering

Anonymous Authors[1]

Abstract

Multimodal vision-language systems incur significant costs from retrieving and processing high-fidelity visual inputs, yet most systems operate at fixed fidelity levels regardless of query requirements. We introduce VOILA, a framework for Value-Of-Information-driven adaptive fidelity selection in Visual Question Answering (VQA) that optimizes *what information to retrieve* before model execution. Given a query, VOILA employs a two-stage calibration pipeline: a gradient-boosted regressor estimates the likelihood of correctness at each fidelity using only question features, while an isotonic calibrator refines these probabilities for reliable decision-making. The system then selects the minimum-cost fidelity that maximizes expected utility by balancing predicted accuracy gains against retrieval and processing costs. We evaluate VOILA across three real-world deployment scenarios—edge-cloud VQA systems, agentic memory architectures, and mission-critical cyber-physical systems—using five datasets (VQA-v2, GQA, TextVQA, LoCoMo, FloodNet) and six vision-language models ranging from 7B to 235B parameters. VOILA consistently achieves 50–60% cost reductions compared to full-resolution processing while maintaining accuracy within 1.5% across diverse query types and model architectures. These results demonstrate that pre-retrieval fidelity selection provides a complementary optimization axis to model routing, enabling efficient multimodal inference under resource constraints.

1. Introduction

Modern multimodal vision-language systems have achieved remarkable accuracy on visual reasoning tasks by scaling model capacity and training data (Alayrac et al., 2022; Li et al., 2023a; Liu et al., 2024a). However, these gains come at substantial cost: multimodal inputs are orders of magnitude larger than text, incurring significantly higher storage, bandwidth, and computational overhead. Recent work has addressed these efficiency challenges through *model selection* and *adaptive routing*—dynamically choosing between lightweight and heavyweight models based on query characteristics (Chen et al., 2023; Ding et al., 2024; Wang & Kuleshov, 2024). While effective at reducing inference costs, these approaches implicitly assume that input context is fixed and freely available, deferring all optimization to the model execution stage.

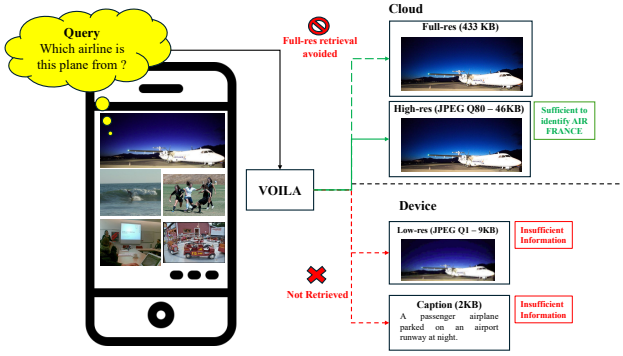


Figure 1. Given a user query, VOILA predicts the minimum visual fidelity required *before retrieval*. For the query "Which airline is this plane from?", low-fidelity on-device representations (caption and JPEG Q1) are insufficient and therefore not retrieved. VOILA directly selects a higher-fidelity cloud representation (JPEG Q80, 46 KB), which preserves the airline logo and is sufficient to answer the query, while avoiding unnecessary access to the full-resolution image (433 KB), yielding a $\sim 9\times$ reduction in network bandwidth.

In practice, this assumption rarely holds. Multimodal inputs often reside across heterogeneous storage tiers with vastly different retrieval latencies and costs. Cloud photo services like Google Photos and iCloud Photos store images at multiple resolutions: on-device caches contain compressed thumbnails and captions, while full-resolution originals reside in cloud storage (Xiao et al., 2023; Zhang et al., 2023). Similarly, enterprise systems use hot storage (e.g., AWS S3 Standard) for frequently accessed data and cold storage (e.g., AWS Glacier) for archival content, with retrieval latencies measured in minutes to hours (Nagaraj et al., 2016). Critically, these retrieval costs are incurred *before* any model processes the input, making post-hoc model routing insuffi-

cient for end-to-end efficiency.

This paper argues that **context selection at the pre-retrieval stage is a distinct and fundamental learning problem**, orthogonal to model routing. Rather than asking *"which model should answer this query?"*, we ask: *"which representation of the available context should be retrieved at all?"* Consider Figure 1: given the query *"Which airline is this plane from?"*, a vision-language model (VLM) cannot answer from a text caption alone (2 KB) but can correctly identify "Air France" from a moderately compressed JPEG (46 KB), avoiding retrieval of the full-resolution image (433 KB)—a 9× bandwidth reduction. The key insight is that **different queries require different minimum input fidelities**, and this variability can be predicted *before* expensive retrieval occurs.

We formalize this setting as a *cost-sensitive information acquisition problem* and introduce **VOILA** (Value-Of-Information guided Latency-Aware fidelity selection), a principled framework for adaptive fidelity selection in multimodal Visual Question Answering (VQA). Given a query, VOILA employs a two-stage pipeline: a gradient-boosted regressor (GBR) estimates, from question-only features, the probability that a given fidelity yields a correct answer; isotonic regression then calibrates these estimates to produce reliable correctness probabilities. Using these calibrated predictions, VOILA selects the fidelity maximizing *expected utility*—anticipated accuracy improvement minus acquisition cost—yielding a one-step value-of-information decision rule whose utility regret is bounded by calibration error.

Unlike existing work on adaptive inference (Zhou et al., 2020; Schuster et al., 2022) and model routing (Shnitzer et al., 2023; Ding et al., 2024), which optimize *how to compute*, VOILA optimizes *what information to acquire*. This distinction is critical in three deployment scenarios: **Edge-Cloud VQA Systems.** On mobile devices running VLMs (Zhang et al., 2024; Chu et al., 2024), captions and low-resolution images are cached on-device while full-resolution images reside in the cloud. Network latency and bandwidth costs dominate total query latency, making pre-retrieval optimization essential. **Agentic Memory Systems.** Long-context conversational agents (Wang et al., 2024; Park et al., 2023) store episodic memories across storage tiers: recent interactions in hot storage, archival memories in cold storage. Retrieving high-fidelity memory traces incurs both latency and monetary costs (Wu et al., 2023). **Mission-Critical Cyber-Physical Systems.** In disaster response scenarios (Rahnemoonfar et al., 2021), autonomous drones must decide which image fidelities to transmit over bandwidth-constrained networks, balancing latency against decision accuracy.

Across these domains, VOILA achieves **50–60% cost reductions** while maintaining accuracy within 1.5% of full-resolution processing on VQA benchmarks (VQA-v2 (Goyal et al., 2017), GQA (Hudson & Manning, 2019), TextVQA (Singh et al., 2019)), long-context memory tasks (LoCoMo (Wang et al., 2024)), and disaster assessment (FloodNet (Rahnemoonfar et al., 2021)). These gains persist across multiple VLMs (Pixtral-12B (AI, 2024), LLaMA-4-Maverick, Qwen-72B-VL (Bai et al., 2023), LLaVA-1.5-7B, Qwen-2.5-VL-7B, and Qwen-3-235B-A22B-instruct), highlighting the robustness of pre-retrieval context selection as a learning primitive.

To our knowledge, VOILA is the first system to formalize and solve pre-retrieval fidelity selection for multimodal AI through value-of-information optimization. Our theoretical analysis shows that VOILA implements a near Bayes-optimal policy when probability estimates are calibrated, with regret bounded by calibration quality. This work establishes **context selection as a first-class learning problem**, complementing model routing and providing a principled foundation for resource-aware decision-making in retrieval-augmented and multimodal systems. We summarize our contributions:

- **Problem formulation:** We formalize pre-retrieval context selection as a cost-sensitive information acquisition problem, orthogonal to model routing (§**??**).
- **Value-of-information framework:** We introduce a principled decision rule that trades off predicted task success against acquisition cost, enabling query-adaptive fidelity selection (§3).
- **VOILA system:** We propose a lightweight, model-agnostic framework combining gradient-boosted regression with isotonic calibration for reliable pre-retrieval fidelity selection (§3).
- **Comprehensive evaluation:** Through experiments spanning three deployment scenarios, six VLMs, and five datasets, we show VOILA consistently improves the accuracy–cost Pareto frontier (§**??**).

2. Background and Motivation

Recent advances in multimodal and retrieval-augmented learning have significantly improved task accuracy by leveraging rich contextual inputs such as images, video frames, and long-term memory traces. However, these gains come with substantial costs: multimodal inputs are orders of magnitude larger than text, incur non-trivial retrieval latency, and often reside across heterogeneous storage tiers. Despite this, most existing systems implicitly assume that all available context can be retrieved upfront, deferring efficiency considerations to model selection or adaptive computation stages.

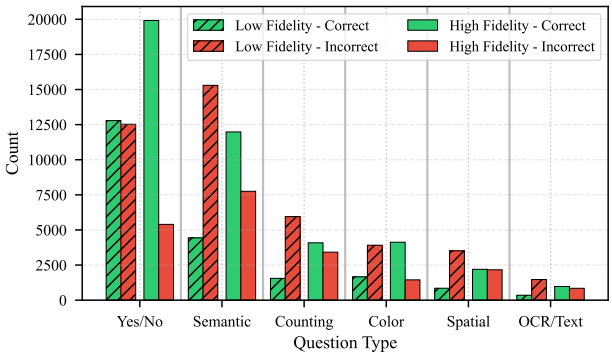


Figure 2. Query-dependent accuracy across input fidelities. Correct and incorrect answer counts are shown by question type for low- and high-fidelity inputs. Low-fidelity representations perform well on semantically coarse queries (e.g., yes/no and some semantic questions) but exhibit high error rates on counting, color, spatial, and OCR-dependent queries. These patterns indicate that different question types require different minimum input fidelities, motivating adaptive, query-dependent fidelity selection.

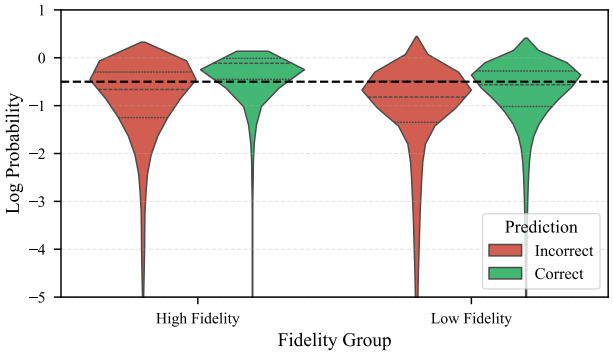


Figure 3. **Model confidence does not reliably indicate context sufficiency.** Violin plots of model log-probabilities for correct and incorrect answers, grouped by low- and high-fidelity inputs. The dashed line indicates a representative confidence threshold that would prevent escalation in a confidence-based system. Incorrect low-fidelity answers frequently exceed this threshold, demonstrating that high model confidence does not imply that the retrieved context contains sufficient information to answer the query correctly.

In this work, we argue that this assumption obscures a more fundamental decision problem: *deciding which context to retrieve at all*. This decision must be made before observing expensive modalities and under uncertainty about their utility for a given query. We motivate this perspective through a series of empirical observations that reveal systematic inefficiencies and failure modes in current multimodal pipelines.

2.1. Problem Setting: Pre-Retrieval Context Selection

We consider a setting in which multiple representations of the same underlying context are available at different fidelities (e.g., captions, compressed images, full-resolution images), each incurring a distinct acquisition cost. The agent must select a single fidelity prior to retrieval in order to balance expected task performance against resource usage. A formal definition of this decision problem, including the Bayes-optimal policy and regret-based evaluation, is provided in Section **??**. Here, we focus on motivating why this formulation captures a critical and underexplored challenge in multimodal learning.

Observation 1: Information Requirements Vary Systematically Across Queries Figure 1 illustrates at an individual-query level that different questions posed over the same image require different minimum input fidelities to be answered correctly. We quantify this effect at scale in Fig. 2, which reports correctness counts across question types for low- and high-fidelity inputs. Low-fidelity representations perform well on semantically coarse questions such as yes/no and some object-identification queries, but fail disproportionately on counting, color, spatial, and OCR-dependent questions that require fine-grained visual evidence. In contrast, high-fidelity inputs substantially improve accuracy on these visually demanding categories while offering limited gains on queries already solvable at low fidelity. Together, these results show that no single fidelity dominates across query types and that information sufficiency is strongly correlated with properties of the query itself, motivating query-adaptive fidelity selection prior to retrieval.

Observation 2: Model Confidence Is an Unreliable Signal for Context Sufficiency: A common heuristic in adaptive multimodal systems is to retrieve low-cost context first and escalate to higher-fidelity inputs only when the model exhibits low confidence, typically measured via token-level log-probabilities. However, we find that model confidence is frequently misaligned with correctness, especially when the retrieved context lacks critical visual information. As shown in Figure 3, incorrect answers produced from low-fidelity inputs often receive log-probability scores comparable to correct answers, frequently exceeding confidence thresholds that would prevent escalation. This leads to systematic failure modes in which confident but incorrect predictions are returned without accessing more informative representations. These results demonstrate that post-retrieval confidence is an unreliable indicator of context sufficiency, motivating the need for pre-retrieval, query-conditioned decision mechanisms.

Observation 3: Model Scaling Does Not Compensate for Insufficient Information: A common response to failures on challenging multimodal queries is to route inputs to larger or more capable vision–language models. However, we find that increasing model capacity does not resolve er-

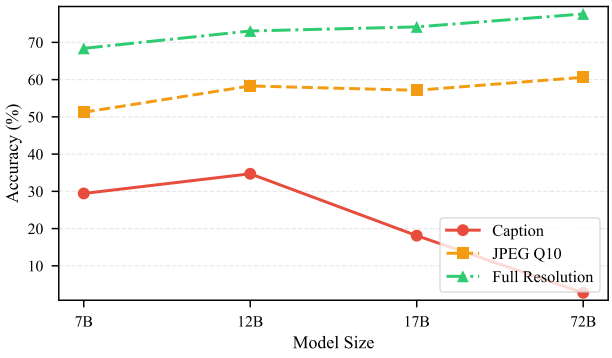


Figure 4. Effect of model capacity under fixed input fidelity. Increasing model size yields limited and non-monotonic accuracy gains when input fidelity is low (caption, JPEG Q10), while full-resolution inputs consistently benefit from scaling. This demonstrates that model capacity alone cannot compensate for insufficient visual information.

rors caused by insufficient or lossy input representations. Figure 4 shows that, when input fidelity is fixed, scaling yields limited and non-monotonic gains: caption-only inputs exhibit early saturation and even degradation at larger model sizes, while moderately informative representations (e.g., JPEG Q10) show only marginal improvements. In contrast, full-resolution inputs consistently benefit from increased capacity. These trends indicate that model failures under low-fidelity inputs stem from missing visual evidence rather than inadequate reasoning ability. Caption-only inputs exhibit early saturation and degradation at larger model sizes, indicating that increased capacity does not compensate for missing visual evidence. Together, these results suggest that information sufficiency is a prerequisite for effective scaling, and that adaptive systems should prioritize decisions about what information to retrieve before selecting which model to apply.

2.2. Observation 4: Calibration Enables Reliable Utility-Aware Decisions

While raw confidence signals are unreliable, we find that probabilistic calibration substantially improves the alignment between predicted success and actual correctness across context fidelities. By learning calibrated estimates of per-fidelity success probabilities from question-only features, it becomes possible to reason about the expected benefit of acquiring additional information before retrieval occurs.

Empirically, calibrated predictors exhibit monotonic and interpretable behavior across fidelities: queries predicted to benefit from higher fidelity indeed show larger accuracy gains, while those predicted to be solvable with low-cost context rarely improve when escalated. This calibration property is essential for enabling principled value-of-information decisions and for providing theoretical guarantees on decision quality. In contrast, uncalibrated predictors and heuristic thresholds fail to provide consistent improvements across datasets and cost regimes.

3. Methodology

We propose **VOILA** (**V**alue-**O**f-**I**nformation for mu**L**timodal **A**ccuracy), a query-adaptive framework for selecting the minimum-cost input fidelity required to answer a multimodal query accurately. VOILA operates entirely at inference time and is model-agnostic, requiring no modification to the underlying vision–language model (VLM).

3.1. Problem Setup

Let (q, x) denote a visual question q paired with an image x. We assume access to a discrete set of input fidelities

$$\mathcal{F} = \{f_1, f_2, \ldots, f_K\},$$

ordered by increasing cost, where lower fidelities correspond to cheaper representations (e.g., captions or compressed images) and higher fidelities correspond to richer representations (e.g., full-resolution images).

Each fidelity $f \in \mathcal{F}$ has an associated cost $c(f) > 0$, capturing retrieval, bandwidth, and compute overhead. Given a fidelity f, the VLM produces an answer $\hat{a}_f$ with token-level log-probabilities.

Our goal is to select, for each query, the fidelity

$$f^\star = \arg\min_{f \in \mathcal{F}} c(f) \quad \text{s.t.} \quad \hat{a}_f \text{ is correct},$$

without access to the ground-truth answer at inference time.

3.2. Fidelity-Specific Supervision

We construct a training corpus by executing the VLM on all fidelities for a set of labeled questions. For each question–fidelity pair, we obtain:

- the model-generated answer $\hat{a}_f$,
- token log-probabilities $\{\log p_f(t)\}$,
- a binary correctness label $y_f \in \{0, 1\}$, computed using a recall-based match between $\hat{a}_f$ and the ground-truth answer.

Answers deemed non-informative (e.g., “unanswerable”, “more information needed”) are excluded from training.

We summarize model confidence using the aggregated log-probability

$$s_f = \sum_{t \in \hat{a}_f} \log p_f(t),$$

which serves as a raw, uncalibrated signal of answer reliability.

3.3. Calibration of Fidelity-Specific Confidence Signals

Raw token log-probabilities produced by vision–language models are known to be poorly calibrated and inconsistent across question types and input fidelities. Directly using these values for decision-making leads to unreliable escalation behavior.

For each fidelity f, we therefore first learn a monotonic calibration function using isotonic regression. Given the aggregated log-probability

$$s_f = \sum_{t \in \hat{a}_f} \log p_f(t),$$

and the corresponding correctness label $y_f \in \{0, 1\}$, we fit an isotonic regressor

$$\psi_f : s_f \to \tilde{p}_f,$$

where $\tilde{p}_f$ represents a calibrated estimate of correctness probability.

This step enforces monotonicity while correcting systematic over- and under-confidence in the raw model outputs.

3.4. Question-Conditioned Success Estimation

While isotonic calibration produces reliable probabilities for *observed* question–fidelity pairs, it does not generalize to unseen queries. To enable prediction without executing the model at every fidelity, we learn a question-conditioned estimator on calibrated signals.

For each fidelity f, we train a Gradient Boosting Regressor (GBR)

$$g_f : \phi(q) \to \tilde{p}_f,$$

where $\phi(q)$ denotes a feature representation of the question, including:

- TF–IDF embeddings of question text,
- lexical features (question length, numeric indicators),
- coarse question-type features (e.g., counting, color, spatial).

The regressor learns to predict the calibrated success probability $\tilde{p}_f$ directly from question features, enabling estimation of fidelity-specific correctness likelihoods without additional model calls.

Algorithm 1 VOILA Inference Procedure

```
Initialize f ← f_1
Compute p̂_f
for f' ∈ F \ {f} in ascending cost do
    Compute p̂_{f'}
    VOI ← (p̂_{f'} − p̂_f) − λc(f')
    if VOI > τ then
        f ← f'
        p̂_f ← p̂_{f'}
    else
        break
    end if
end for
return answer from fidelity f
```

We denote the resulting prediction as

$$\hat{p}_f = g_f(\phi(q)).$$

3.5. Value-of-Information Criterion

At inference time, VOILA performs incremental fidelity selection starting from the cheapest representation. Let f_i be the current fidelity with calibrated success probability $\hat{p}_{f_i}$.

For a candidate higher fidelity f_j, we define the Value of Information (VOI) as:

$$\mathrm{VOI}(f_j \mid f_i) = (\hat{p}_{f_j} - \hat{p}_{f_i}) - \lambda \cdot c(f_j),$$

where λ controls the trade-off between accuracy gains and cost.

Intuitively, VOI measures whether the expected improvement in correctness justifies the additional resource expenditure.

4. Evaluation

We evaluate VOILA across a diverse set of datasets, model architectures, and deployment scenarios to assess its effectiveness as a pre-retrieval fidelity selection mechanism for cost-aware multimodal inference. Our evaluation is designed to answer four key questions: (i) whether VOILA consistently improves the accuracy–cost trade-off relative to fixed-fidelity baselines, (ii) whether these gains persist across model scale and dataset characteristics and (iii) which components of VOILA are critical to its performance.

Models: We evaluate six state-of-the-art vision–language models spanning a wide range of parameter scales and architectural designs: *Pixtral-12B*, *LLaMA-4-Maverick (17B×12E)*, *Qwen2-VL-72B*, *LLaVA-1.5-7B*, *Qwen2.5-VL-7B*, and *Qwen-3-VL-235B-A22B*. These models range from

lightweight 7B parameter models to a 235B-parameter mixture-of-experts model, enabling us to study whether fidelity selection remains beneficial under aggressive model scaling. All models are used in a zero-shot or instruction-following setting.

Datasets: We conduct experiments on five datasets covering complementary query distributions and deployment scenarios: **VQA-v2** and **GQA** serve as standard in-domain VQA benchmarks with diverse question types; **TextVQA** emphasizes OCR-dependent and text-centric queries; **FloodNet** represents a mission-critical, bandwidth-constrained cyber-physical setting involving aerial imagery for disaster assessment; and **LoCoMo** evaluates long-horizon, agentic memory retrieval with generated VQA-style questions grounded in conversational context. Together, these datasets allow us to test VOILA under varying degrees of visual complexity, linguistic structure, and system constraints.

Input Fidelities: For each image, we construct a discrete set of input fidelities ordered by increasing cost: textual captions, 32×32 resized thumbnails, JPEG images at low (Q1) and medium (Q10) quality, and full-resolution images. Fixed-fidelity baselines always operate at a single fidelity, while VOILA selects among these options on a per-query basis using its value-of-information criterion.

Cost Estimation Methodology: Because VOILA optimizes *what information to acquire* prior to model execution, our cost model measures the overhead incurred *before inference*, rather than model-internal computation. Each input fidelity is assigned a normalized acquisition cost $c(f)$ intended to approximate the relative system cost of retrieving that representation in a tiered deployment, combining data transfer volume, storage access tier, and retrieval latency. Costs are derived by measuring the average on-disk size of each fidelity and mapping them to canonical storage tiers: captions and low-resolution thumbnails are assumed to be cached locally (edge or hot storage), compressed JPEG images are retrieved from warm storage, and full-resolution images reside in cold storage. Higher-cost tiers incur larger penalties to reflect increased bandwidth usage and access latency. This abstraction captures key deployment regimes motivating our formulation, including edge–cloud VQA systems (bandwidth-dominated), agentic memory systems (latency- and retrieval-dominated), and mission-critical cyber-physical settings (transmission-constrained). Finally, all costs are normalized such that full-resolution images have cost 120, with other fidelities scaled relative to this reference. Unless otherwise specified, we use a fixed edge–cloud cost profile for consistency across experiments.

Ablation Studies: We conduct a comprehensive ablation analysis to isolate the contribution of each component of VOILA. First, we vary the question-conditioned predictor used to estimate per-fidelity success probabilities, comparing Gradient Boosted Regression (GBR) against logistic regression, ridge regression, and multilayer perceptrons. This evaluates the importance of modeling non-linear relationships between question features and fidelity-specific success. Second, we study the effect of probabilistic calibration by replacing isotonic regression with uncalibrated predictors, measuring changes in both cost and calibration error. This ablation tests the hypothesis that reliable probability estimates are necessary for stable value-of-information decisions. All ablations use the same adaptive VOI routing policy and are evaluated across six models and two representative datasets (VQA-v2 and GQA), allowing us to disentangle improvements due to better probability estimation from those due to routing logic alone.

Evaluation Metrics: We report task accuracy and average cost per query for each method. Performance is assessed by comparing the accuracy–cost achieved by VOILA against fixed-fidelity baselines. For calibration quality, we additionally report Brier score and Expected Calibration Error (ECE) in the ablation studies.

5. Results and Discussions

VOILA consistently matches near–full-resolution accuracy at dramatically lower cost: Across four standard VQA benchmarks (VQA-v2, GQA, TextVQA, FloodNet) and six vision–language models, VOILA consistently achieves near–full-resolution accuracy while reducing average retrieval cost by 50–60 %.

Fixed-fidelity baselines are Pareto-dominated No fixed input fidelity is optimal across datasets or models: captions systematically underperform on visually demanding queries, while always using high-resolution inputs incurs unnecessary cost on a large fraction of queries.

Gains persist under distribution shift and scale: VOILA's cost–accuracy gains persist across model sizes from 7B to 235B, indicating that adaptive fidelity selection is complementary to, and not replaceable by, model scaling.

Out-of-Distribution Generalization Holds When trained on VQA-v2 and evaluated on GQA, VOILA maintains its accuracy–cost tradeoff, demonstrating that fidelity requirements are query-driven rather than dataset-specific.

VOILA Extends Beyond Static Benchmarks to Agentic Memory Systems.: Despite its distinct setup, LoCoMo demonstrates that VOILA's value-of-information principle extends beyond static VQA to agentic memory systems, achieving comparable accuracy to full-resolution retrieval at over 70 % lower cost.

VOILA Enables Cost-Aware Perception in IoT / CPS Systems: On FloodNet, which models an IoT cyber-physical setting where aerial images are captured by drones and trans-

mitted to a base station, VOILA significantly reduces communication and retrieval costs while preserving decision-critical accuracy.

Calibration Enables Reliable Value-of-Information Routing: An ablation study across six vision–language models and two datasets shows that VOILA's performance depends critically on accurate probability calibration rather than the specific choice of predictor architecture. While different regressors (GBR, logistic regression, ridge, MLP) yield similar raw accuracy, calibrated variants consistently achieve lower cost at comparable accuracy by making more reliable escalation decisions. In contrast, uncalibrated predictors exhibit higher utility variance and unnecessary escalation. These results confirm that calibration—not model complexity—is the key enabler of stable, cost-aware value-of-information routing.

- Distribution of fidelity selection 2x2 figure.
- OOD distribution results.
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6. Related Work

Optimizing the trade-off between accuracy and computational cost in Visual Question Answering (VQA) has prompted a range of innovations in adaptive routing, modular system design, and fidelity-aware inference. Farazi et al. (Farazi et al., 2020) systematically studied the cost of multimodal fusion in VQA pipelines, showing that simplifying or bypassing this expensive stage can preserve competitive accuracy while greatly reducing computation. (Weng et al., 2024) introduced RACC (Retrieval-Augmented MLLMs with Compressed Contexts), demonstrating that compressing retrieved knowledge for KB-VQA can reduce inference latency by 22–60% without sacrificing state-of-the-art accuracy. Li et al. (Li et al., 2023b) showed how multi-level semantic fusion of question-only features enables rich reasoning without visual input, aligning with VOILA's emphasis on question-derived routing. Similarly, Wang et al. (Wang et al., 2021) explored latent modeling of captions and answer types during training, improving test-time accuracy without relying on intermediate model outputs—a conceptual precursor to VOILA's pre-retrieval decision-making.

From a broader methodological perspective, VOILA draws on work in budgeted information acquisition and value-of-information (VoI) theory. Wickens's early research on expected value theory in human visual attention allocation (Wickens et al., 2001) and its applications in resource-constrained decision-making (e.g., healthcare) (Heath et al., 2024) demonstrate how selectively acquiring additional inputs can optimize task performance. Kossen et al. (Kossen et al., 2022) formulated the Active Acquisition for Multimodal Temporal Data (A2MT) task, where agents choose which costly input modalities to acquire in a temporal sequence—conceptually similar to VOILA's modality acquisition from question-only cues. Bilgic and Getoor (Bilgic & Getoor, 2014) introduced the VOILA framework (Value Of Information Lattice) for optimal, cost-sensitive feature subset selection using probabilistic dependencies, further reinforcing the theoretical foundation for selective acquisition. However, these approaches typically assume access to intermediate model signals or are tailored to unimodal or tabular domains, whereas VOILA operates entirely on pre-retrieval, question-only features in a multimodal reasoning context.

Adaptive inference methods, such as early-exit architectures and dynamic resource allocation (Liu et al., 2024b; Wang et al., 2025), also aim to tailor computational effort to input complexity, but generally focus on optimizing computation within a single model architecture. In contrast, VOILA performs cross-fidelity routing—deciding between captions, compressed images, and full-resolution images—under an explicit VoI objective that balances predicted accuracy against modality acquisition cost. This distinction makes VOILA well suited for integration with retrieval-based or agentic memory systems, where the cost of acquiring additional modalities can vary significantly.

To our knowledge, VOILA is the first system to combine supervised calibration via Gradient Boosted Regressors (GBRs) with a VoI-driven routing framework for fidelity selection in VQA. VOILA estimates the likelihood of correctness for each modality using only question-derived features and calibrates these predictions independently of model outputs or ground-truth labels at test time—enabling principled, interpretable, and budget-aware routing. While GBRs have been applied in cost-sensitive learning (Zheng et al., 2017; Peter et al., 2017; Kärkkäinen et al., 2019) and VoI has been used extensively in decision-critical domains (Heath et al., 2024; Wickens et al., 2001), VOILA is the first deployable system to unite these components for adaptive multimodal reasoning under strict computational or budget constraints.

References

AI, M. Pixtral 12b. `https://mistral.ai/news/pixtral-12b/`, 2024.

Alayrac, J.-B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al. Flamingo: a visual language model for few-shot learning. In *Advances in Neural Information Processing Systems*, volume 35, pp. 23716–23736, 2022.

Bai, J., Bai, S., Yang, S., Wang, S., Tan, S., Wang, P., Lin, J., Zhou, C., and Zhou, J. Qwen-vl: A versatile vision-

Table 1. **Complete Comparison Table (VQA-v2, GQA, FloodNet, TextVQA).** Accuracy (%) and average cost for each method across datasets and models. FloodNet uses a deployment-specific transmission cost model reflecting bandwidth-constrained disaster-response settings. **VOI** rows are bolded.

Model	Method	VQA-v2		GQA		FloodNet		TextVQA	
		Acc	Cost	Acc	Cost	Acc	Cost	Acc	Cost
Pixtral-12B-2409	Caption	49.83	10	42.45	10	47.84	5	51.31	10
	VOI	**68.67**	**49.90**	**67.37**	**50.18**	**79.48**	**38.40**	**75.92**	**47.35**
	JPEG Q10	69.21	70	69.81	70	75.75	80	74.82	70
	JPEG Q1	57.27	50	60.53	50	55.70	30	52.28	50
	Resize 32x32	33.80	20	38.10	20	52.05	12	19.57	20
	Original	74.06	120	73.52	120	81.51	120	79.29	120
Llama-4-Maverick (17Bx12E)	Caption	45.98	10	34.93	10	45.01	5	51.70	10
	VOI	**64.61**	**47.67**	**60.87**	**45.76**	**69.14**	**43.60**	**64.52**	**49.43**
	JPEG Q10	65.99	70	63.17	70	65.23	80	65.48	70
	JPEG Q1	50.45	50	52.76	50	60.41	30	45.11	50
	Resize 32x32	39.42	20	43.92	20	55.73	12	25.11	20
	Original	67.37	120	63.56	120	70.35	120	71.56	120
Qwen2-VL-72B-Instruct	Caption	40.21	10	32.34	10	56.48	5	58.19	10
	VOI	**73.01**	**52.75**	**71.03**	**49.86**	**75.51**	**33.30**	**88.65**	**56.74**
	JPEG Q10	74.27	70	71.57	70	78.07	80	86.84	70
	JPEG Q1	60.49	50	62.68	50	79.84	30	58.96	50
	Resize 32x32	43.90	20	49.63	20	76.85	12	19.27	20
	Original	77.16	120	74.35	120	76.08	120	93.11	120
llava-1.5-7b-hf	Caption	58.01	10	64.09	10	51.05	5	47.80	10
	VOI	**68.07**	**42.33**	**71.18**	**29.52**	**80.12**	**44.80**	**64.85**	**52.33**
	JPEG Q10	70.21	70	74.11	70	73.64	80	63.25	70
	JPEG Q1	58.25	50	63.51	50	70.76	30	49.50	50
	Resize 32x32	58.99	20	67.37	20	66.22	12	21.50	20
	Original	72.80	120	75.77	120	82.61	120	66.45	120
Qwen2.5-VL-7B-Instruct	Caption	36.65	10	41.28	10	48.06	5	43.48	10
	VOI	**72.01**	**62.98**	**62.92**	**65.57**	**78.17**	**38.30**	**72.90**	**55.74**
	JPEG Q10	72.10	70	60.14	70	61.24	80	70.26	70
	JPEG Q1	56.26	50	48.66	50	56.81	30	41.74	50
	Resize 32x32	48.38	20	46.70	20	50.06	12	11.35	20
	Original	77.98	120	67.22	120	77.19	120	73.87	120
Qwen3-VL-235B-A22B-Instruct	Caption	57.36	10	28.97	10	52.99	5	59.05	10
	VOI	**74.57**	**57.19**	**68.28**	**49.12**	**80.62**	**36.20**	**94.10**	**56.87**
	JPEG Q10	72.58	70	70.40	70	82.95	80	93.65	70
	JPEG Q1	62.62	50	62.58	50	81.06	30	69.85	50
	Resize 32x32	50.21	20	54.08	20	72.31	12	22.00	20
	Original	79.01	120	72.98	120	83.39	120	97.15	120

Table 2. **VOILA ablations (VQA-v2 / GQA) across 6 models.** Each cell reports Acc / Cost / Brier . Savings is omitted. All runs use adaptive VOI routing; fixed-fidelity baselines are excluded. *(GBR cells are aligned to the main VOI table for Acc/Cost.)*

Model	Ablation	VQA-v2 (Acc / Cost / Brier)				GQA (Acc / Cost / Brier)			
		GBR	LogReg	Ridge	MLP	GBR	LogReg	Ridge	MLP
Pixtral	Predictor	0.6867 / 49.90 / 0.23	0.70 / 45.07 / 0.21	0.70 / 48.00 / 0.21	0.70 / 52.15 / 0.21	0.6737 / 50.18 / 0.21	0.69 / 55.68 / 0.20	0.68 / 56.18 / 0.20	0.68 / 56.18 / 0.20
Pixtral	Calibration	0.6867 / 49.90 / 0.23	0.70 / 46.52 / 0.23	0.68 / 31.04 / 0.22	0.70 / 49.18 / 0.21	0.6737 / 50.18 / 0.21	0.67 / 52.38 / 0.21	0.66 / 39.03 / 0.22	0.67 / 57.02 / 0.20
LLaMA-4	Predictor	0.6461 / 47.67 / 0.23	0.66 / 59.48 / 0.22	0.65 / 57.28 / 0.22	0.66 / 60.43 / 0.22	0.6087 / 45.76 / 0.23	0.62 / 57.63 / 0.22	0.62 / 54.32 / 0.22	0.63 / 58.94 / 0.22
LLaMA-4	Calibration	0.6461 / 47.67 / 0.23	0.65 / 53.91 / 0.23	0.65 / 45.20 / 0.24	0.65 / 54.80 / 0.22	0.6087 / 45.76 / 0.23	0.61 / 50.03 / 0.23	0.60 / 37.66 / 0.24	0.62 / 53.26 / 0.22
Qwen-72B-VL	Predictor	0.7301 / 52.75 / 0.21	0.74 / 64.08 / 0.20	0.74 / 64.25 / 0.20	0.73 / 62.03 / 0.20	0.7103 / 49.86 / 0.21	0.72 / 54.58 / 0.20	0.72 / 56.90 / 0.20	0.71 / 53.31 / 0.20
Qwen-72B-VL	Calibration	0.7301 / 52.75 / 0.21	0.73 / 58.84 / 0.21	0.73 / 54.43 / 0.21	0.73 / 60.49 / 0.20	0.7103 / 49.86 / 0.21	0.71 / 54.97 / 0.21	0.71 / 47.28 / 0.21	0.72 / 57.10 / 0.20
LLaVA	Predictor	0.6807 / 42.33 / 0.22	0.69 / 52.91 / 0.21	0.69 / 51.63 / 0.21	0.68 / 47.15 / 0.21	0.7118 / 29.52 / 0.20	0.72 / 21.44 / 0.19	0.71 / 22.64 / 0.19	0.72 / 25.68 / 0.19
LLaVA	Calibration	0.6807 / 42.33 / 0.22	0.68 / 48.36 / 0.22	0.70 / 66.16 / 0.22	0.68 / 52.38 / 0.21	0.7118 / 29.52 / 0.20	0.71 / 24.98 / 0.20	0.71 / 15.66 / 0.20	0.72 / 23.50 / 0.19
Qwen-VL-7B	Predictor	0.7201 / 62.98 / 0.22	0.74 / 80.25 / 0.21	0.74 / 82.50 / 0.21	0.74 / 76.99 / 0.21	0.6292 / 65.57 / 0.22	0.66 / 59.36 / 0.21	0.66 / 58.84 / 0.21	0.65 / 58.27 / 0.21
Qwen-VL-7B	Calibration	0.7201 / 62.98 / 0.22	0.73 / 69.77 / 0.22	0.75 / 87.36 / 0.22	0.74 / 78.43 / 0.21	0.6292 / 65.57 / 0.22	0.63 / 47.39 / 0.22	0.67 / 77.01 / 0.24	0.65 / 57.78 / 0.21
Qwen-3-VL-235B	Predictor	0.7457 / 57.19 / 0.20	0.76 / 73.23 / 0.20	0.76 / 70.59 / 0.20	0.75 / 69.69 / 0.20	0.6828 / 49.12 / 0.21	0.69 / 35.24 / 0.20	0.70 / 36.50 / 0.20	0.70 / 36.61 / 0.20
Qwen-3-VL-235B	Calibration	0.7457 / 57.19 / 0.20	0.75 / 68.03 / 0.20	0.77 / 71.22 / 0.21	0.76 / 73.17 / 0.20	0.6828 / 49.12 / 0.21	0.69 / 35.34 / 0.21	0.68 / 23.95 / 0.21	0.69 / 36.18 / 0.20

language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 2023.

Bilgic, M. and Getoor, L. Value of information lattice: Exploiting probabilistic independence for effective feature subset acquisition. *ArXiv*, 2014.

Chen, L., Zaharia, M., and Zou, J. Frugalgpt: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.

Chu, X., Qiao, L., Lin, X., Xu, S., Yang, Y., Hu, Y., Wei, F., Zhang, X., Han, B., et al. Mobilevlm: A fast, reproducible and strong vision language assistant for mobile devices. *arXiv preprint arXiv:2312.16886*, 2024.

Ding, D., Mallick, A., Wang, C., Sim, R., Mukherjee, S., Ramos, V., Awadallah, A., Raghunathan, A., et al. Hybrid llm: Cost-efficient and quality-aware query routing. *arXiv preprint arXiv:2404.14618*, 2024.

Farazi, M., Khan, S. H., and Barnes, N. Accuracy vs. complexity: A trade-off in visual question answering models. *Pattern Recognition*, 120:108106, 2020. doi: 10.1016/J.PATCOG.2021.108106.

Goyal, Y., Khot, T., Summers-Stay, D., Batra, D., and Parikh, D. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 6904–6913, 2017.

Heath, A. et al. Applications of value of information in resource-constrained decision making. *Decision Support Systems*, 2024.

Hudson, D. A. and Manning, C. D. Gqa: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6700–6709, 2019.

Kossen, J., Cangea, C., Vértes, E., Jaegle, A., Patraucean, V., Ktena, I., Tomasev, N., and Belgrave, D. Active acquisition for multimodal temporal data: A challenging decision-making task. *ArXiv*, 2022.

Kärkkäinen, K. et al. Cost-sensitive feature-value acquisition. *ArXiv*, 2019.

Li, J., Li, D., Savarese, S., and Hoi, S. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International Conference on Machine Learning*, pp. 19730–19742. PMLR, 2023a.

Li, S., Gong, C., Zhu, Y., Luo, C., Hong, Y., and Lv, X. Context-aware multi-level question embedding fusion for visual question answering. *Information Fusion*, 102: 102000, 2023b. doi: 10.1016/j.inffus.2023.102000.

Liu, H., Li, C., Wu, Q., and Lee, Y. J. Visual instruction tuning. *Advances in Neural Information Processing Systems*, 36, 2024a.

Liu, H., Wang, B., Sun, Y., Li, X., Hu, Y., and Yin, B. Vig: Visual information-guided knowledge-based visual question answering. In *CSCWD*, pp. 1086–1091, 2024b. doi: 10.1109/CSCWD61410.2024.10580527.

Nagaraj, K., Bharadia, D., Mao, H., Chinchali, S., Alizadeh, M., and Katti, S. Cooling the data center: Optimization of cooling infrastructure and workload placement. In *Proceedings of the USENIX Annual Technical Conference*, pp. 283–296, 2016.

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., and Bernstein, M. S. Generative agents: Interactive simulacra of human behavior. pp. 1–22, 2023.

Peter, S., Diego, F., Hamprecht, F., and Nadler, B. Cost efficient gradient boosting. *ArXiv*, pp. 1551–1561, 2017. URL `https://consensus.app/papers/cost-efficient-gradient-boosting-peter-diego/bb6361f1fa8c5120a0121e4b026de98e/?utm_source=chatgpt`.

Rahnemoonfar, M., Chowdhury, T., Sarkar, A., Varshney, D., Yari, M., and Murphy, R. R. Floodnet: A high resolution aerial imagery dataset for post flood scene understanding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pp. 1486–1495, 2021.

Schuster, T., Fisch, A., Gupta, J., Dehghani, M., Bahri, D., Tran, V. Q., Tay, Y., and Metzler, D. Confident adaptive language modeling. In *Advances in Neural Information Processing Systems*, volume 35, pp. 17456–17472, 2022.

Shnitzer, T., Ou, A., Soares, C., et al. Large language model routing with benchmark datasets. *arXiv preprint arXiv:2309.15789*, 2023.

Singh, A., Natarajan, V., Shah, M., Jiang, Y., Chen, X., Batra, D., Parikh, D., and Rohrbach, M. Towards vqa models that can read. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 8317–8326, 2019.

Wang, F., Peng, L., Cao, S., Yang, Z., Zhang, R., and An, G. Vcf: An effective vision-centric framework for visual question answering. *Neurocomputing*, 2025. doi: 10.1016/j.neucom.2025.129536.

Wang, I. and Kuleshov, V. Routellm: Learning to route llms with preference data. *arXiv preprint arXiv:2406.18665*, 2024.

Wang, Z., Miao, Y., and Specia, L. Latent variable models for visual question answering. In *ICCVW*, pp. 3137–3141, 2021. doi: 10.1109/ICCVW54120.2021.00351.

Wang, Z., Feng, A., Pollard, C., Williams, A., and Callison-Burch, C. Locomo: Learning to correlate long-term conversational memory. *arXiv preprint arXiv:2404.05719*, 2024.

Weng, W., Zhu, J., Zhang, H., Meng, X., Zhang, R., and Yuan, C. Learning to compress contexts for efficient knowledge-based visual question answering. *ArXiv*, 2024. doi: 10.48550/arXiv.2409.07331.

Wickens, C., Helleberg, J., Goh, J., Xu, X., and Horrey, W. Pilot task management: Testing an attentional expected value model of visual scanning. *ARL*, 2001.

Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Awadallah, A., et al. Autogen: Enabling next-gen llm applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*, 2023.

Xiao, W., Zhang, L., and Chen, M. Storage optimization in mobile devices: A survey. In *IEEE International Conference on Mobile Cloud Computing, Services, and Engineering*, pp. 112–119. IEEE, 2023.

Zhang, B., Zhou, D., Zhang, Y., Wang, J., Wang, Y., and Zhao, X. Tinyllava: A framework of small-scale large multimodal models. *arXiv preprint arXiv:2402.14289*, 2024.

Zhang, Y., Liu, W., and Chen, X. Hierarchical storage management for mobile visual data. *IEEE Transactions on Mobile Computing*, 22(8):4567–4580, 2023.

Zheng, C., Kasprowicz, C. G., and Saunders, C. Customized routing optimization based on gradient boost regressor model. *ArXiv*, abs/1710.11118, 2017. URL `https://consensus.app/papers/customized-routing-optimization-based-on-gradient-df5e7f667eeb5776b381b2820feebcf8/?utm_source=chatgpt`.

Zhou, W., Xu, C., Ge, T., McAuley, J., Xu, K., and Wei, F. Bert loses patience: Fast and robust inference with early exit. *Advances in Neural Information Processing Systems*, 33:18330–18341, 2020.

A. Appendix

You can have as much text here as you want. The main body must be at most 8 pages long. For the final version, one more page can be added. If you want, you can use an appendix like this one.

The `\onecolumn` command above can be kept in place if you prefer a one-column appendix, or can be removed if you prefer a two-column appendix. Apart from this possible change, the style (font size, spacing, margins, page numbering, etc.) should be kept the same as the main body.